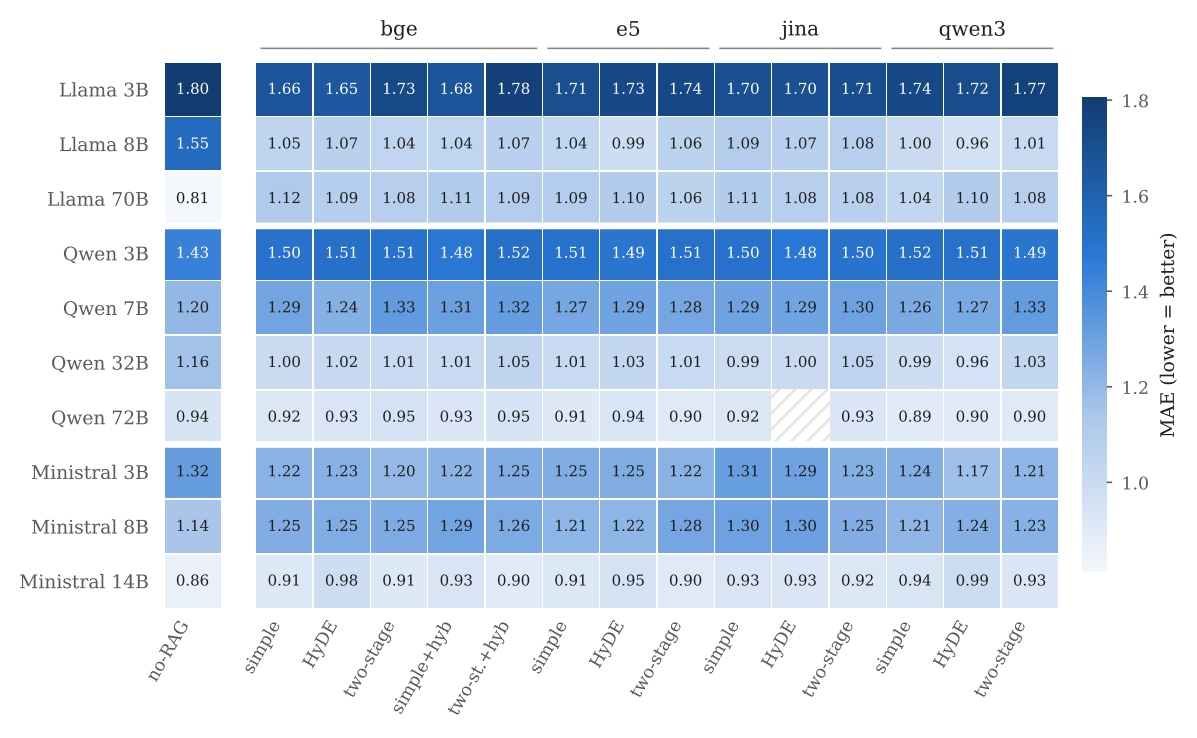
bge
e5
jina
qwen3
Llama 3B 1.80 1.66 1.65 1.73 1.68 1.78 1.71 1.73 1.74 1.70 1.70 1.71 1.74 1.72 1.77
Llama 8B 1.55 1.05 1.07 1.04 1.04 1.07 1.04 0.99 1.06 1.09 1.07 1.08 1.00 0.96 1.01
Llama 70B 0.81 1.12 1.09 1.08 1.11 1.09 1.09 1.10 1.06 1.11 1.08 1.08 1.04 1.10 1.08
Qwen 3B 1.43 1.50 1.51 1.51 1.48 1.52 1.51 1.49 1.51 1.50 1.48 1.50 1.52 1.51 1.49
Qwen 7B 1.20 1.29 1.24 1.33 1.31 1.32 1.27 1.29 1.28 1.29 1.29 1.30 1.26 1.27 1.33
Qwen 32B 1.16 1.00 1.02 1.01 1.01 1.05 1.01 1.03 1.01 0.99 1.00 1.05 0.99 0.96 1.03
Qwen 72B 0.94 0.92 0.93 0.95 0.93 0.95 0.91 0.94 0.90 0.92 0.93 0.89 0.90 0.90
Ministral 3B 1.32 1.22 1.23 1.20 1.22 1.25 1.25 1.25 1.22 1.31 1.29 1.23 1.24 1.17 1.21
Ministral 8B 1.14 1.25 1.25 1.25 1.29 1.26 1.21 1.22 1.28 1.30 1.30 1.25 1.21 1.24 1.23
Ministral 14B 0.86 0.91 0.98 0.91 0.93 0.90 0.91 0.95 0.90 0.93 0.93 0.92 0.94 0.99 0.93
no-RAG
simple
HyDE
two-stage
simple+hyb
two-st.+hyb
simple
HyDE
two-stage
simple
HyDE
two-stage
simple
HyDE
two-stage
1.8
1.6
1.4
1.2
1.0
MAE (lower = better)